

Geometric Origin of 10^{11} in the Gravitational Constant

An Empirical Observation (v2.3 — corrected)

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Abstract

This note presents an empirical geometric observation: the factor 10^{11} appearing in the gravitational constant G is not an arbitrary decimal power. It arises as an approximation from four turns of a 3D golden spiral with an additional micro-correction (breathing term) of $1/720$. The approximation reaches 0.020% precision (IRON level), not an exact identity.

1. The Spiral Block

The fundamental structural block is defined using only the golden ratio φ and the integer 31:

$$S = \varphi^6 \times 31 \approx 556.27242921$$

The logarithmic inclination per spiral turn:

$$\kappa = \log_{10}(S) \approx 2.745\,287\,535$$

Four full turns give:

$$4\kappa \approx 10.981\,150\,141$$

Target: $\log_{10}(10^{11}) = 11.000\,000\,000$. Gap: $0.018\,849\,859$ ($\approx 0.171\,36\%$ of 11).

2. Micro-step (Breathing Term)

One full turn consists of 12 micro-steps, each contributing the basic breathing unit $1/720$:

$$12 \times \frac{1}{720} = \frac{1}{60} \approx 0.016666667$$

After correction:

$$10.981150141 + 0.016666667 = 10.997816808$$

Residual gap: $11.000000 - 10.997816808 = \mathbf{0.002183192}$ ($\approx 0.020\%$, **IRON**).

This is an approximation, not an exact identity.

3. Resonant Relation

$$\log_{10}(10^{11}) \approx 4 \times \log_{10}(\varphi^6 \times 31) + \frac{12}{720}$$

Holds to 0.020% precision.

4. Equivalent Forms of G_{Koch}

$$G_{\text{Koch}} = \frac{31^2 \times \sqrt{5}}{\varphi^{12} \times 10^{11}} \quad (1)$$

Since $\varphi^{12}/\sqrt{5} = 144 + 1/720$, we have $\varphi^{12} = (144 + 1/720) \cdot \sqrt{5}$. Substituting into form (1):

$$G_{\text{Koch}} = \frac{31^2 \sqrt{5}}{(144 + 1/720) \cdot \sqrt{5} \times 10^{11}}$$

The $\sqrt{5}$ cancels exactly, giving:

$$G_{\text{Koch}} = \frac{31^2}{(144 + 1/720) \times 10^{11}} \quad (2)$$

$\sqrt{5}$ cancels algebraically. This form contains only integers and $1/720$. Both forms are equivalent and give:

$$G_{\text{Koch}} \approx 6.67354674496 \dots \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$$

Gap from CODATA 2018: -0.011% (IRON).

5. The Residual $1/720$ — Three Independent Domains

Domain	Formula	Gap	Status
Geometry	$\varphi^{12}/\sqrt{5} - 144 = 1/720 + \mathcal{O}(10^{-8})$	$\sim 10^{-8}\%$	ABS.IRON
Nuclear	$m_n/m_p = 1 + 1/720$	0.001%	ABS.IRON
EM	$\alpha^{-1} = 144 - 7 - 1/720$	0.027%	STEEL

One residual appears consistently across independent domains.

6. What This Is Not

This is an empirical geometric observation. No dynamical model is claimed. The relation $4\kappa + 12/720 \approx 11$ holds to 0.020% (IRON), not exact. No CODATA values were used in the derivation.

Full precision reference:

$$\begin{aligned} G_{\text{Koch}} = & 6.67354674496161812375919977179042360791321745487151 \\ & 55655761367183204332559409890102447913340675769273 \\ & \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2} \end{aligned}$$

See also:

Kochenov Geometric Identity for G (Zenodo, April 2026)

The Integer 31 as a Discrete Geometric Node in Kochenov G (Zenodo, May 2026)

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